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Disaster City Digital Twin: A vision for integrating artificial and human intelligence for disaster management

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ABSTRACT

This paper presents a vision for a Disaster City Digital Twin paradigm that can: (i) enable interdisciplinary convergence in the field of crisis informatics and information and communication technology (ICT) in disaster management; (ii) integrate artificial intelligence (AI) algorithms and approaches to improve situation assessment, decision making, and coordination among various stakeholders; and (iii) enable increased visibility into network dynamics of complex disaster management and humanitarian actions. The number of humanitarian relief actions is growing due to the increased frequency of natural and man-made crises. Various streams of research across different disciplines have focused on ICT and AI solutions for enhancing disaster management processes. However, most of the existing research is fragmented without a common vision towards a converging paradigm. Recognizing this, this paper presents the Disaster City Digital Twin as a unifying paradigm. The four main components of the proposed Digital Twin paradigm include: multi-data sensing for data collection, data integration and analytics, multi-actor game-theoretic decision making, and dynamic network analysis. For each component, the current state of the art related to AI methods and approaches are examined and gaps are identified.

1. Introduction

The objective of this paper is to present a vision for a Disaster City Digital Twin as an integrated paradigm for converging different streams of research related to artificial intelligence in disasters and crisis informatics. With the increased frequency of natural and man-made disasters and crises, the need for disaster responses and humanitarian relief actions is growing rapidly (Jackson et al., 2018). The need for improving the effectiveness and efficiency of disaster management has been widely recognized, especially in the mission and policies of the United Nations (United Nations, 2019). One important area to improve disaster response and emergency management is the employment of information and communication technologies. To this end, various streams of research across different disciplines such as information science, computer science and social science have devoted efforts to developing and employing information and communication technology (ICT) and artificial intelligence (AI) approaches for enhancing disaster response and emergency management processes (Fan & Mostafavi, 2019). For example, researchers in information and social science have focused on disaster coordination networks to improve the efficiency of information spread and communications (Hamadache, Seridi-Bouchelaghem, & Farah, 2016; Pearson, Pearce, & Kingham, 2013). In addition, AI techniques including natural language processing (e.g., event extraction, named entity recognition, and speech recognition) and image processing (e.g., image segmentation and denoising) have been utilized for leveraging intelligence learned by machines from disaster situational data (Nweke, Teh, Mujtaba, & Al-garadi, 2019). Researchers from computer science primarily focus on improving the accuracy and capabilities of AI techniques to capture the dynamic situations in disasters (Nam & Pardo, 2011; Pohl, Bouchachia, & Hellwagner, 2012). For example, the Center for Integrated Emergency Management (CIEM) in Norway has carried out research to develop AI and ICT techniques such as information sharing and crowdsourcing tools for crisis responders. For instance, the CIEM proposed an agenda for design of disaster-specific ICT to enhance information sharing (Gjøsæter, Radianti, & Chen, 2018). Specifically, the agenda includes data reliability examination (Lazreg et al., 2018), data analytics for extracting situational information (Sharma, Granmo, & Goodwin, 2018), and information sharing and communication (Alsalamah et al., 2018).

Given the growing literature in the interdisciplinary field of disaster/crisis information processing and AI in disaster management,

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there is a dire need for interdisciplinary convergence towards a unified vision. Recognizing this gap, this paper aims to provide a vision for Disaster City Digital Twin as a unifying paradigm for interdisciplinary convergence. Digital twin is an emerging paradigm which aims to build a digital version of the physical and human environment with the situational information (Mohammadi & Taylor, 2017). The digital twin paradigm has been envisioned in the management and operation of smart cities (Min, Lu, Liu, Su, & Wang, 2019; Neirotti, De Marco, Cagliano, Mangano, & Scorrano, 2014). The disaster context is a special scenario in smart cities, in which the situation evolves rapidly and becomes complex. A digital twin paradigm in a city affected by disasters could provide numerous benefits for enhanced situation assessment, decision making, coordination, and resource allocation. In a Disaster City Digital Twin, spatiotemporal dynamics of disaster regions and humanitarian actions are integrated into an analytics platform fusing datasets from crowdsourcing tools and agencies. Through fusion, learning, and exchange of spatiotemporal information with various relief actors (enabled through data integration and visualization) and the virtual coordination, the digital twin of a disaster city can provide a vision for convergence of various streams of research related to ICT and AI in disaster response and emergency management.

Integrating the ICT and AI techniques from research conducted across different disciplines into a digital twin paradigm would require four main components in a Disaster City Digital Twin. The four components of the proposed Digital Twin paradigm are: (1) multi-data sensing for data collection, (2) data integration and analytics, (3) multi-actor game-theoretic decision making, and (4) dynamic network analysis (see Fig. 1). The first component focuses on AI technologies and methods for situational data collection from multiple sources in disasters and humanitarian emergencies. Disaster situations and humanitarian crises are often characterized as data-starved due to constraint resources for data gathering and analysis. However, the advances in AI has brought opportunities to address this challenge and gather, store, and analyze various types of data related to a disaster city. In particular, AI-enabled remote sensing, social sensing and crowdsourcing technologies are discussed as important elements of the Digital Twin for near real-time gathering and analysis of disaster and crisis situations. The second component of the Digital Twin is geared towards the employment of AI in integration of heterogeneous data in order to draw important insights needed by responders and relief actors. Specifically, we discuss the challenges of dealing with different types of information (e.g., social media posts, volunteer and crowdsourced data, aerial photos, maps, reports, and news articles) and AI solutions (such as knowledge graph and network embedding) to implement machine learning on heterogeneous data to inform disaster management and relief actions. The third component of the Digital Twin includes a Serious Game Learning Environment to enable multi-actor decision making and networked coordination. In this component, we examine the employment of AI for improving disaster response training and network-centric coordination based on approaches for multi-actor gaming scenarios. The fourth component of the envisioned Digital Twin involves dynamic network analysis capturing the interactions among various types of networks such as actors and information networks (e.g., who coordinates with who; who does what relief tasks; and who needs what information) for performance assessment of disaster management and humanitarian actions. In this component of the Digital Twin, we discuss the employment of AI to examine missing links in the network and to record temporal information in order to improve the efficiency of disaster management efforts.

Through collection, analytics, training, and exchange of situational information with the humanitarian actors (enabled through data integration and visualization) and the virtual coordination (enabled by serious gaming and dynamic network analysis), the digital twin of a disaster city and its human users become smarter over time and gain

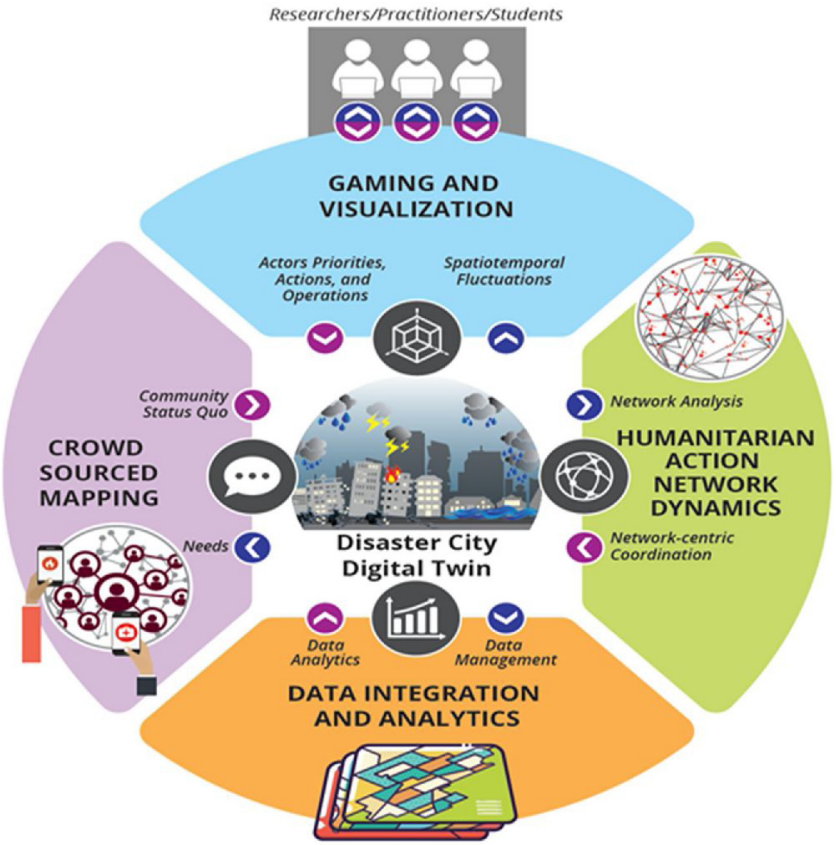


Fig. 1. Overview of the Disaster City Digital Twin Paradigm.

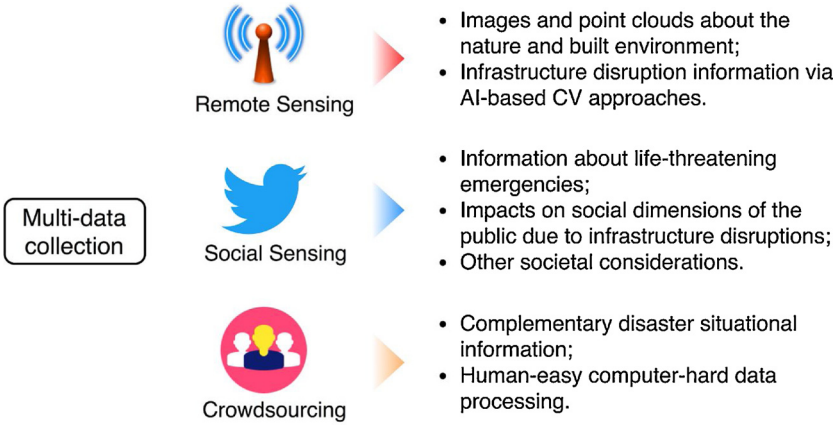


Fig. 2. Multi-data sensing for data collection.

predictive insights into the planning and response operations in humanitarian actions. Here, humanitarian and emergency response actors are broadly defined as robot-human teaming. The following sections present each of the four components of the Disaster City Digital Twin paradigm. For each component, we discuss clusters of recently published papers and the-state-of-the-art techniques that can be integrated into this Digital Twin paradigm and discuss the opportunities for effective employment of AI towards the realization of a vision for the Disaster City Digital Twin.

2. Multi-data sensing for data collection

The information for the Disaster City Digital Twin paradigm is mainly obtained from three sources: remote sensing, social sensing, and crowdsourced data collection (see Fig. 2). Remote sensing includes satellite and UAV-based aerial imagery (Akter & Wamba, 2017), which is primarily used for capturing the anomaly and changes of natural and built environment before, during, and after disasters. Satellite imagery can provide multiple types of information for disaster management and response. For example, for tropical cyclone disasters (i.e. hurricanes and Typhoons), satellite imagery can produce data about land use, land cover, storm surge height, and cyclone wind speed. UAV-based aerial imagery and video captured via unmanned aerial vehicles (UAVs) are playing an increasingly important role in disaster response, due to its efficiency for rapid mapping (Gomez & Purdie, 2016). In addition, aerial imagery can produce detailed 3D point clouds of the environment and infrastructure in two ways. First, a CCD camera on UAVs can generate mutually overlapped 2D images (or video frames) to be transferred to 3D point clouds through Structure from Motion (SfM) algorithm (Westoby, Brasington, Glasser, Hambrey, & Reynolds, 2012). Second, some UAVs carry Light Detection and Ranging (LiDAR) sensors can directly produce 3D point clouds (Hoque, Phinn, & Roelfsema, 2017). Another advantage of UAV-based aerial imagery over satellite images or piloted aerial imagery is that they can deploy efficiently in disaster environments with minimum supporting infrastructure facilities (e.g. satellites and airports) (Yu et al., 2018). Therefore, UAVs are now an emerging technology to inspect the impacts of disasters, such as small-scale changes and cracks of buildings infrastructures due to earthquakes, hurricanes, or landslides (Haworth & Bruce, 2015).

Different artificial intelligence techniques can contribute to improving the deployment of remote sensing in disasters (Walker, 2016). In the Digital Twin Paradigm, images and point cloud models obtained from remote sensing could be automatically analyzed through different AI techniques. First, image segmentation can divide remote sensing images into regions according to the semantic content (e.g. urban areas, farm, forest, etc.), which is critical in acquiring land use and land cover information. Second, object detection and tracking can be used to identify infrastructure components (e.g. bridges, road networks) to improve the efficiency of assessing the damages to critical infrastructure. Change detection can identify the differences between two remote-sensing images that were spatially registered, which is a convent way to detect deformation of nature or built environment due to floods, droughts, or earthquakes (Cooner, Shao, & Campbell, 2016).

Social sensing based on Natural Language Processing (NLP) and data mining techniques is another emerging methodology to extract and analyze social media information for evaluating disaster phenomena (Atefeh & Khreich, 2013). Social media provide rich information source about disaster situations. Data analytics on publicly-available social media contents can support: 1) detecting or predicting critical events (e.g., power outage or need for shelter); and 2) identifying underlying patterns of social media users to understand human behaviors in disasters (Arthur, Boulton, Shotton, & Williams, 2018). Compared to traditional data collection approaches such as survey and field interviews, social sensing has several significant advantages (Arthur et al., 2018; Zhang, Fan, Yao, Hu, & Mostafavi, 2019). First, social media data analytics mainly rely on the content generated by the public instead of disaster responders, which do not put a burden on the human resources for disaster management. Second, social media analytics can provide near real-time transitions of disaster situations, which can potentially improve the decision-making process for efficient and effective disaster response.

Despite the usage and benefits of social sensing for disaster response, the potential of social sensing for situation awareness in disasters has not been fully achieved. For example, posts on social media platforms contain rich content about infrastructure disruptions (e.g., power outage or contaminated water) as well as people's reaction, adjustment, and needs related to the service disruptions (Castillo, 2016). Therefore, analyzing the spatiotemporal patterns and correlation between infrastructure and societal impacts based on social media posts can provide valuable insights for infrastructure agencies and researchers to understand the interactions between human and physical environment and develop planning and response strategies. In addition, existing social sensing approaches suffer from a weakness related to fake information. Achieving real-time detection and elimination of fake information on social media requires further achievements in many areas about AI in social sensing, such as rumor dissemination pattern, characteristics of fake posts, and recognition of rumor-debunking behaviors.

Crowdsourcing is another trending data collection and analysis method with cost-effectiveness and time-efficiency during disasters. AI techniques can potentially improve the performance of crowdsourcing techniques in disaster response and emergency management by providing high quality and structured data for training and testing machine learning algorithms. Crowdsourcing refers to a family of data collection and/or decision-making approaches based on aggregating the thoughts of agents for higher decision quality than that based on the information

provided by individuals (Arganda-Carreras et al., 2015). In the domain of disaster management, crowdsourcing approaches generally have two roles: collecting data and processing data (Ofli et al., 2016). Crowdsourcing for data collection allows people to proactively report first-hand, real-time information about disaster situations to online platforms. Crowdsourcing for data processing refers to approaches in which people implement human-easy and computer-difficult tasks (e.g. labeling images, adding coordinates, tagging reports with categories, etc.) to generate structured, high quality, interpreted data for decision making or machine learning. One typical application of crowdsourcing for data processing is segmenting and labeling damaged buildings in remote sensing imageries and social media texts. Hence, crowdsourcing is not only a way to collect high-quality disaster situational information, but also contribute to processing remote and social sensing data by providing labeled images and social media posts for training machine-learning models in the Digital Twin paradigm.

The growing use of crowdsourcing technologies could also improve the performance of machine learning algorithms (e.g., Human-in-the-loop (HITL) machine learning) for classifying data and extracting insights in the Digital Twin. HITL machine learning could improve the effectiveness and efficiency of machine learning algorithms by combining human and machine intelligence. In HITL machine learning, human interactions are involved in both training and testing stages to reduce an exponential search space (Holzinger et al., 2016) and improve learning accuracy (Xin et al., 2018). In the training stage, humans provide ground-truth labels for the training data in order to make accurate predictions; and in the testing stage, humans can give quick feedback to correct the results where the algorithms are not confident about a judgment. Hence, HITL machine learning algorithms can improve the performance of different components of the Disaster City Digital Twin paradigm. For example, disaster situation information (e.g., texts, images, and videos) from social media and crowdsourced platforms can be labeled (e.g., determine the level of building damages and the needs of relief actions) by human volunteers in preparing the training set and correct training errors to improve deep learning algorithms used for damage classification.

Hence, remote sensing, social sensing and crowdsourcing can provide multimodal information related to the natural and built environment, infrastructure disruptions, life-threatening emergencies, and impact on social dimensions. This information is pivotal in mapping and visualizing evolving stations in the Disaster City Digital Twin paradigm. Remote sensing provides imagery information about the natural and built environment for detecting information disruption via AI-based computer vision algorithms. Social sensing, on the other hand, will provide information about life-threatening emergencies witnessed by people and societal considerations of the public during the disasters, such as the impact on social dimensions due to infrastructure disruptions. Finally, crowdsourcing techniques enable collecting complementary disaster situational information that is difficult to capture by remote of social sensing in a targeted manner. Furthermore, crowdsourcing can integrate human knowledge into AI approaches by generating training datasets of labeled remote sensing imageries or social media posts. Together, remote sensing, social sensing, and crowdsourcing information processed via the state-of-the-art AI approaches can support the efficiency and effectiveness of data collection and processing for the Disaster City Digital Twin paradigm.

3. Data integration and analytics

Being aware of the dynamic situation in disasters is significantly important for effective disaster preparedness, response and recovery (Fan & Mostafavi, 2018; Hashem et al., 2016). One of the key components in understanding the disaster situation is the exploitation of situational information in disasters (Fan, Mostafavi, Gupta, & Zhang, 2018). As mentioned in the previous section, available techniques for gathering information from different types of detectors (a.k.a. "modality") have increased over the past decade (Dalla Mura et al., 2015). The complex nature of disaster situations is difficult to be characterized by information from a single modality, and hence, requires integration of multiple modalities to provide complete knowledge about disaster situations and phenomena (Lahat, Jutten, Lahat, Lahat, & Jutten, 2015). However, the increased availability of multimodal data raises challenges in data integration and analytics due to the lack of theoretical and algorithmic strategies in representation of heterogeneous data (Nweke et al., 2019). The primary issues related to dealing with heterogeneous data in disasters are twofold: first, data generated from various sources: social media, remote sensors, and other sources are stored in multiple different data structures (von Lubitz, Beakley, & Patricelli, 2008); second, different types of data exist in a single modality, which increases the complexity of data characterization (Lahat et al., 2015). For example, social media as an emerging modality includes texts, images, videos, geolocations, and hyperlinks describing their emotions, activities or built environment disruptions during disasters (Alam, Imran, & Ofli, 2017; Alam, Ofli, & Imran, 2018). In addition, remote sensing technologies such as synthetic aperture radar (SAR) and LiDAR can acquire highly precise thermal images and large periodic numeric data for condition monitoring and damage assessment (Dalla Mura et al., 2015). The heterogeneity of data formats, timeframes, and semantics makes it very challenging to efficiently and accurately understand the spatiotemporal fluctuations in disasters and humanitarian actions in the Digital Twin paradigm.

To deal with the challenges arising from multimodal data, existing studies have attempted to develop new models and algorithms to integrate multi-source data and then extract insights from the data content for the optimization of multisensory systems, understanding of biomedical functionality, and environment observations (see Fig. 3) (de Albuquerque, Herfort, Brenning, & Zipf, 2015). One of the proposed strategies for data integration is constructing a knowledge graph in which entities are represented as nodes and relationships between entities are captured as edges; both the nodes and edges in a knowledge graph are high-dimensional (Paulheim, 2015). Knowledge graph is based on Resource Description Framework (RDF) which has an < Subject, Predicate, Object > model, where Subject and Object represent entities and Predicate represents relationships (Imran, Castillo, Diaz, & Vieweg, 2014). In particular, for disaster conditions, the key idea for a unified knowledge graph is to define the central entities in disaster cities and then project the entities in the high-dimensional space. Well-defined entities enable filtering relevant information from multimodal data and preparing precise representation of the information in knowledge graphs.

In addition, modalities are inherently interrelated and could provide complementary information since they may describe the same entities through different features. Thus, multiple modalities can potentially help us achieve better representation, in the sense that each modality brings to the whole some types of added value that cannot be deduced or obtained from any of the other modalities (Lahat et al., 2015). Lack of consideration of the relationships among modalities may induce information redundancy or loss. Hence, identifying and assessing the complementarity (or non-linear relationships) among multimodal data is important to obtaining complete knowledge and reduce information redundancy or loss. However, little of the existing work has examined the complementarity in multimodal data, especially in the context of disasters. Therefore, one important area for further integration of AI for data fusion in the knowledge graph is to measure similarities of the information, distinguish complementary components, and learn projection functions for data representation through different feature spaces (Bengio, Courville, & Vincent, 2013). The multimodal information without redundancy would enhance the accuracy and efficiency of situation assessment in the Disaster City Digital Twin.

In addition, knowledge graphs, as heterogeneous information networks (HINs), can be used as input to various analyses in the Digital Twin paradigm to understand the relationships among different

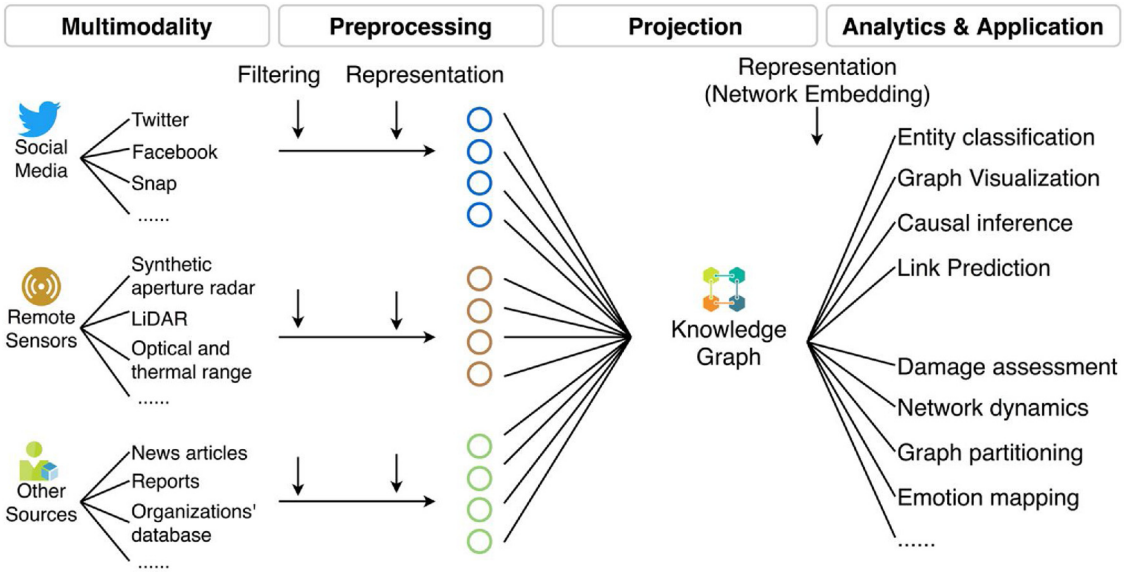


Fig. 3. Data Integration and Analytics in the Disaster City Digital Twin.

entities. This capability may hold the key to inference and prediction of the dynamic fluctuations and situation in disasters, and further lead to an improvement of disaster response and community resilience. In particular, to leverage inference and learning capabilities of deep learning (DL) algorithms, there is a need to transform the graph or network structures into a low-dimensional vector space. Note that recent work (Schlichtkrull et al., 2018) has made progress on graph neural networks but currently the most effective DL algorithms are based on vector input. Network embedding, as a representation approach, can transform the vertices of knowledge graphs and HINs into low-dimensional (embedded space) vectors, such that both geometrical properties of the networks and attribute information about the nodes can be preserved in the vectors (Huang, Li, & Hu, 2017). Machine learning algorithms then operate in the embedded space (Ying et al., 2018). With this use of embedded space of HINs, the Disaster City Digital Twin can have knowledge-rich deep learning to enhance the assistive abilities to:

- Process and understand human speech;
- Process and classify sounds (natural and man-made);
- Classify the content of images, sensor/IoT data, and remote-sensing data;
- Generate dense captions for video streams, identifying what activities are in the video;
- Predict future sequences of events based on past event sequences;
- Translate human speech and written languages;
- Generate action sequences based on observed states toward a goal in an optimal manner; and
- Imagine things unseen and future scenarios.

The existing methodologies for solving and implementing network embedding, including hierarchical graph aggregation and graph neural networks, offer a powerful paradigm for node classification, visualization and relation inference in HINs. Furthermore, representation vectors can generate 2D visualizations of graphs, discover hidden structures, classify entities involved, and identify different roles of different entity classes. The latent relationships can better model and depict the interdependencies and interactions among different entities such as critical infrastructure and humanitarian response organizations in the Digital Twin paradigm.

4. Multi-actor game-theoretic decision making

In addition to supporting situation assessment, another main function of the Disaster City Digital Twin is improving networked coordination among heterogeneous actors involved in disaster response and emergency management. Effective and efficient disaster management and relief operations not only rely on the precise situational information, but also require networked coordination and decision making among different relief actors. To enhance collaborative decision making, training relief actors with simulated disaster situations and learning their behaviors for policy and guideline making are needed. This component of the Disaster City Digital Twin aims to enable conducting interactive virtual simulations to identify and examine coordination strategies among different actors involved in disaster response. Accordingly, the Disaster Response Digital Twin can achieve multi-actor training for collaborative operation and decision making in different disaster scenarios.

Serious Gaming Learning and Coordination Environment (SGLCE) would be an important part of the Digital Twin to allow decision-aware serious gaming, coordination, and visualization of disaster response actions. Serious Game is the use of scientific game-engine simulations of real-world phenomena to solve a problem. In SGLCE, actors and decision-makers can engage in decision-making scenarios and collaborate with other actors and decision-makers. Actors can access data about situations, ask what-if queries, analyze scenarios, visualize scenarios, and go through a vigorous and open process of deciding (Roy et al., 2010). What makes SGLCE an important element in the Digital Twin is its ability to simulate the coordination, information flow, and dynamic interactions in disaster response and emergency management. In addition, the visualization component of SGLCE includes the integration of spatiotemporal data and game visuals onto the real world with GPS coordinates and GIS capabilities. GIS-aware agents associated with humanitarian actors can be modeled with a polygonal boundary corresponding to their area of authority/operations.

The decisions of an actor, made to achieve a specific task, can benefit or punish the actor and influence their neighbors (Jackson & Zenou, 2015). For example, an actor may need to decide on an ensemble of tasks that will produce an outcome in the future with the presence of uncertainties. Given the tasks to arrive at the outcome, in the digital twin, these tasks can be formulated as a graph of tasks and dependencies culminating in the attainment of the outcome (Sohn, Oh, & Lee, 2018). Fig. 4 shows an illustration of the task network for

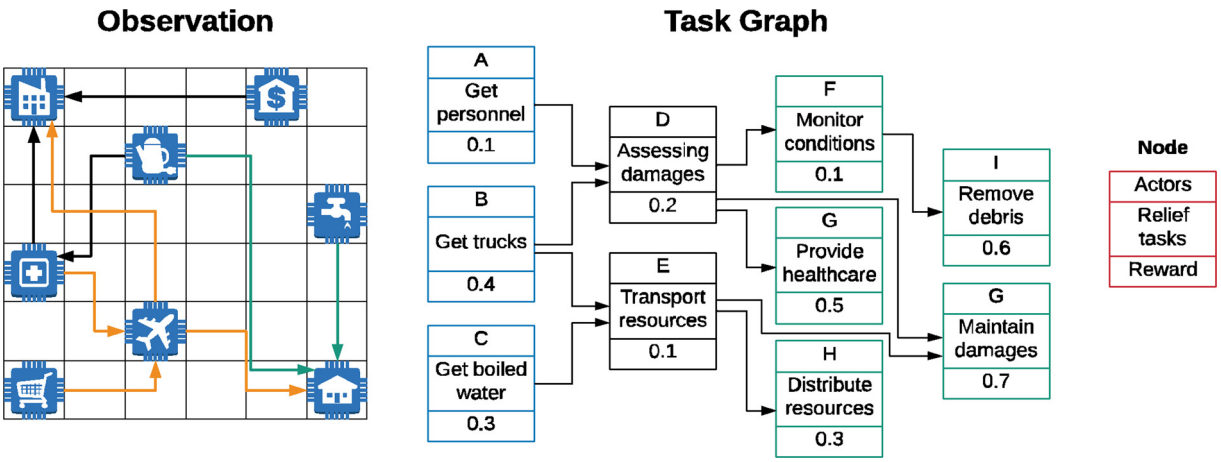


Fig. 4. An illustration of task graph and agents' trajectory in Disaster City Digital Twin.

disaster management. The decision-maker (actor) needs to find the optimal path to, say, remove debris (I). The task graph (as shown) denotes which tasks depend on subtasks (e.g., Task D depends on Task A and Task B) and the resource competition (e.g., Task D competes with Task E to get trucks of Task B). Then, dynamic programming, deep learning and other techniques (e.g., hierarchical reinforcement learning (Nachum, Gu, Lee, & Levine, 2018) and risk analysis (De La Maza, Davis, Gonzalez, & Azevedo, 2019)) can be applied to find optimal task sequences in the task graph. The analysis of potential paths elucidates the trade-off between exploitation and exploration, and between short-term and long-term reward pursuits. The task graph itself can be learned from empirical data. This is important in the fluid situation where tasks change dynamically. The envisioned Digital Twin will include intelligent task graph generator from HIN and task (networked) sequence optimizer for disaster management that are more flexible and smarter than standard operating procedure currently in use (Abdeyazdan, Parsa, & Rahmani, 2013).

Decision making in the real world, however, needs to take into account other actors. There may be not just two opposing actors, but many diverse actors/stakeholders with complementary goals as well. This illustration is for a single actor, but the formalism straightforwardly extends to multiple actors in both collaboration and competitive (game) environment. This calls for gamified and game-theoretic approach, for which the Digital Twin is envisioned to include deep learning techniques similar to DeepMind's pioneering line of winning automated game players such as AlphaGo and Alpha Zero (Wang et al., 2016). However, in the Disaster City Digital twin, the challenge is going from perfect-information board games such as Go to imperfect-information, partially observable, continuous real-time, and irreversible consequential world in disaster cities. In the Digital Twin paradigm, there are a multitude of actors who have different interests and knowledge and they all make their decisions either alone or in coalition. Thus, the outcome is based on a confounded multitude of factors and uncertainties. Alpha Zero with its policy and value network and Monte-Carlo tree search (MCTS) provides a way to model a decision-making process of an actor in the face of adversaries. The policy and value network learn from self-play experiences, while MCTS explores options. The game-theoretic approach can have more than two actors and have schedules for actors to make and execute their decisions, with mixed and hidden human and AI players. The Digital Twin endows game experiments with a gamified scientific visualization, e.g., using Unreal Engine game engine. The visual aspects are essential in the Digital Twin to enable intensive engagement among algorithms and human actors. Integrating these techniques to record, analyze and evaluate the multi-actor interactions in the Disaster City Digital Twin enhances the capability of this system and its human users.

5. Dynamic network analysis

The Disaster City Digital Twin enables increasing the visibility of network dynamics among heterogeneous relief actors, tasks, information, infrastructure and resources in disasters to better understand and enhance the efficiency of response and emergency management. The efficiency of disaster response and emergency management processes is critical to reducing the adverse impacts of disasters on communities. Capturing and addressing inefficiencies such as un-optimized resource distribution and limited information sharing, is essential to improving the performance of humanitarian actions and disaster management. Existing studies have shown that inefficiencies tend to arise from dynamic and complex relationships among different entities and the dynamic variation of their statuses (e.g., emergence or disappearance of new relationships) in disasters (Fan & Mostafavi, 2019). Besides the coordination among multiple relief actors (discussed in Section 4), there are fragmentations (such as the capability of actors to get situational information, the availability of resources for relief tasks, and the functionality of infrastructure for task execution) that can lead to inefficiencies in disaster response and management processes. Hence, an important function of the Digital Twin is to increase the visibility of complex relationships among the entities to improve performance assessment in emergency response.

Existing studies have highlighted the merits of network approaches in analyzing heterogeneous entities and complex relationships (Mcculloh, Lospinoso, & Carley, 2010; Shiau, Dwivedi, & Yang, 2017). Meta-network analysis approach is one of the emerging network approaches, which can abstract various types of entities (e.g., relief actors, tasks, information, and resources) and their interactions. The utility of meta-network approach in analyzing the coordination and communication among various types of entities in disasters have been examined and demonstrated in existing studies (Fan, Zhang et al., 2018; Zhu & Mostafavi, 2018). Meta-networks can include, but are not limited to social networks (who knows who), knowledge networks (who knows what), tasks networks (who does what), and needs networks (what knowledge is needed to do a task) (see Fig. 5) (Altman, Carley, & Reminga, 2017). The Digital Twin with meta-network analysis component could enable evaluating the interactions among different entities, identifying inefficiencies induced by their complex relationships, and providing solutions to enhance the performance of disaster management processes (Misra, Goswami, Mondal, & Jana, 2017). In addition, temporal information recorded in the Disaster City Digital Twin enables monitoring, analyzing and predicting the dynamic structures of the networks and the potential effects on the efficiency of relief and response actions.

Integrating AI methods into dynamic meta-network analysis can improve the capabilities of the Disaster City Digital Twin to enhance the

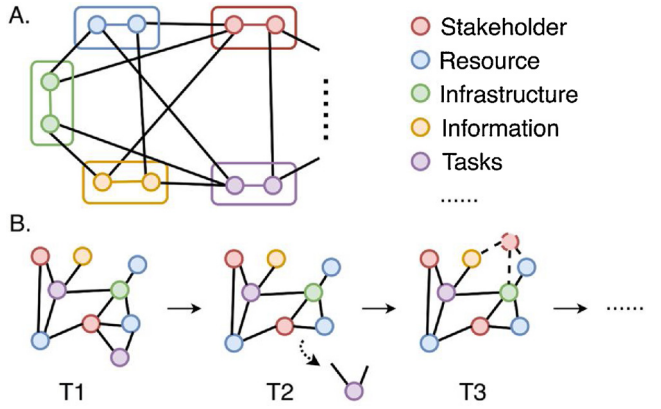


Fig. 5. An illustration of Disaster City Meta-Network (A) and its Dynamic Process (B). T1, T2, and T3 represent timestamps.

performance of disaster management. One important capability in this component of the Digital Twin is examining the criticality of entities and the influence of the entities' statuses on the functioning of other entities in the network. For example, removal of certain entities such as relief actors may cause some other entities inactive (e.g., some tasks cannot be conducted, or some information may be blocked) in the network. Bayesian network analysis methods can learn such effects from the interactions among entities using empirical data, and further infer the status of the nodes at following timestamps based on the changes of other entities (Plomaritis, Costas, & Ferreira, 2017). By doing so, the Digital Twin can provide a deep understanding of the importance of the entities and develop strategies to strengthen the capacities of the critical entities to enhance the overall performance of disaster response and emergency management processes.

Another capability of this component in the Digital Twin is predicting the latent connections and interactions among different entities to enhance the synergetic operation and management. It is often the case that different relief actors work on their relief tasks in isolation. This may raise inefficiencies such as increased costs for resources transport and incomplete information about the disaster situation. Capturing the latent connections among the entities based on their shared resources, similar tasks, and empirical experiences and knowledge can enable the collaboration and cooperation among these entities to enhance the performance of disaster relief (Hu, Wang, Peng, Liang, & Du, 2017). Techniques such as common neighbors detection, preferential attachment, hierarchical structural decomposition have been proposed to predict the connections among entities in a network (Dong et al., 2018). The identification of the potential connections would enable the Disaster City Digital Twin to capture opportunities for cooperation among the entities and improve their performance through synergetic operations.

6. Discussion and recommendations for future research

In this paper, different streams of research related to employing AI in disaster management have been discussed to establish an integrative vision for the Disaster City Digital Twin paradigm. The proposed Disaster City Digital Twin vision offers important contributions and implications for research and practice of AI and city management in disasters to inform future research.

Table 1 summarizes the state of the art, gaps and opportunities, and functions that AI techniques can play in the Disaster City Digital Twin paradigm. To achieve the vision for the Disaster City Digital Twin paradigm, there is a real need for integration of different streams of research to create AI-based converging solutions related to each component. The key functions of each components of the Disaster City Digital Twin paradigm could be summarized as follows:

Table 1
Disaster City Digital Twin Components and AI Techniques.

Components	Disaster phases	State of the art	Gaps/opportunities	Functions that AI can improve
Multi-data sensing for data collection	Preparedness; response; and recovery	Satellite and UAV-based aerial imagery; information spread; and HITL ML.	Image segmentation; rumor detection; and data labeling.	Rapidly collect situational information; estimate the scale of disasters; and map the damages across the affected areas.
Data integration and analytics	Response; recovery; and mitigation	Knowledge graphs; network embedding; and entity classification.	Projection function for dimensionality reduction and redundancy reduction.	Obtain a comprehensive understanding of the disaster situation; and identify relationships among involved entities.
Multi-actor game-theoretic decision making	Response; and recovery	AlphaZero; Monte-Carlo tree search.	Model decision making process; and task graph generator.	Identify a confounded multitude of factors and uncertainties derived from multiple actors; and evaluate the interactions among actors.
Dynamic network analysis	Preparedness; response; and recovery	Meta-networks; and link prediction.	Infer relations among actors; and predict dynamics of networks.	Assess the performance of disaster management processes; and identify key entities.

- The first component of the digital twin paradigm mainly focuses on collecting and visualizing different types of data using AI techniques in disasters (Stieglitz, Mirbabaie, Ross, & Neuberger, 2018). Despite the advances in employing AI to rapidly map and estimate disaster impacts (Gao, Barbier, & Goolsby, 2011) based on textual (Ragini, Anand, & Bhaskar, 2018) and imagery information (Argandа-Carreras et al., 2015), the quality and resolution of the information still require improvement.
- The second component of the digital twin leverages the capabilities of data analytics (i.e., knowledge graph (Paulheim, 2015) and network embedding (Huang, Li, & Hu, 2017)) for fusing multimodal data to provide a comprehensive understanding of disaster situations (Lazreg et al., 2018). To overcome existing challenges such as information redundancy, new methods are needed to determine effective projection functions.
- The third component of the digital twin is geared towards conducting AI-based multi-actor and game-theoretic simulations to enable engaging multiple relief actors for collective decision making (Dwivedi et al., 2019). Learning and modeling the decision making processes of human actors in the digital twin requires the development of task graphs (Abdeyazdan et al., 2013), as well as MXNet deep learning frameworks (Chen et al., 2015).
- The last component of the digital twin focuses on analyzing the performance of disaster management using network learning and prediction techniques (Fan & Mostafavi, 2019). The spatiotemporal fluctuations in the cyber-human-physical systems of cities in disasters can be represented with multiplex dynamic models and be subsequently analyzed to provide timely assessment of the performance of disaster management processes and systems through various network measure and missing link identification techniques (Ben Yahia, Eljaoued, Bellamine Ben Saoud, & Colomo-Palacios, 2019).

6.1. Theoretical contributions and implications

The vision presented in this paper contributes to a theoretical understanding of the convergence of various streams of disaster informatics and AI research and the advances in the interdisciplinary field. In particular, the proposed vision for digital twin contributes to establishing a common vision for interdisciplinary researchers across various fields. The gaps identified related to each component of the Disaster City Digital Twin paradigm could inform “core AI” research about techniques and methods required for smart and intelligent disaster Management. In addition, by providing a converging vision and a shared mental model for researchers from different fields (e.g., information sciences, civil engineering, social sciences, computer and data sciences), the Disaster City Digital Twin could expedite use-inspired AI solutions and innovations in the field of disaster informatics.

6.2. Implications for practice

From a practical perspective, the proposed Disaster City Digital Twin paradigm could provide important capabilities for real-time monitoring, data analytics, and scenario simulation. First, the real-time monitoring capability of the digital twin is achieved through collection of situational data and examination of the spatiotemporal fluctuations related to disaster events such as community disruptions and damages (Fan, Jiang, & Mostafavi, 2020). With automated techniques and tools for continuous data gathering through remote and social sensing, the digital twin enables visualizing human activities, damages, and relief needs across time and space. Accordingly, the public officials, victims, first responders, and volunteers can capture the changes of the situation by tracking the temporal information in disaster-affected areas. Second, the data analytics capability in the digital twin could enable better optimization of disaster management and emergency response operations among various actors. In particular, since the Disaster City Digital Twin paradigm compiles, fuses, and analyzes the data in an integrative and consistent way, the flow of information among different components of the disaster management systems and processes can be significantly enhanced (Ismagilova, Hughes, Dwivedi, & Raman, 2019). This capability would enable a better collaboration and cooperation among different departments and agencies in response to disasters through learning and exchanges of situational information. Finally, the digital twin enables the scenario-play and simulation capabilities for training and planning purposes and to improve cooperation among various and fair allocation of resources. By doing so, the digital twin of a disaster city and its human users could become smarter and more resilient to extreme events over time (Hashem et al., 2016).

As different streams of research converge, the Disaster City Digital Twin can assist the disaster management and emergency response processes. For example, when a hurricane lands in a city, heavy rainfall would result in flooding over metropolitan districts. Roads would be congested or closed, and a great number of residents would lose power and water supplies. Satellites and drones are used to collect image data. Crowdsourcing platforms and social media are employed to gather information about the needs of affected people and the damages in the affected areas. The Digital Twin then aggregates all types of data from these data sources and cleans the data through rumor detection and noise removal using HITL machine learning approaches. Built upon the cleaned situational data, a knowledge graph with entities involved in disasters would be created in the Digital Twin to accurately capture the disaster situation as events unfold. Relief actors have been trained in simulated scenarios with similar knowledge graphs so that they can rapidly coordinate with each other to develop the strategies and conduct relief tasks. Through the interactions among different entities recorded, the Digital Twin would identify the entities with a high demand of capacities and develop improvement strategies in time to enhance the performance of the critical entities as well as the efficiency of the whole network. As envisioned in this simple use case, the Disaster City Digital Twin will augment the capability of communities to cope with future disasters. This vision could only be realized through a convergence of interdisciplinary research streams related to ICT and AI in disaster response and emergency management (Duan, Edwards, & Dwivedi, 2019).

6.3. Limitations and future research direction

The capabilities specified for each component of the Disaster City Digital Twin paradigm could not be fully achieved without further advancements in AI techniques. There are important limitations that need to be addressed in future research to bridge the gaps among different streams of studies for enhancing disaster management processes through integrative ICT and AI techniques. First, as the massive situational data is generated in disasters, the data sensing and collection component enable rapid collection of a variety of data using satellite, crowdsourcing platforms and social media. However, these data tend to have noises, rumors, and false information, which impede the extraction of insightful information for the awareness of disaster situations (Stieglitz, Mirbabaie, Ross, & Neuberger, 2018). The capabilities of this paradigm would be sensitive to the conditions that the situational data is doubtable. Integrating machine learning and crowdsourcing techniques to empower the Digital Twin with the robust capability of data labeling and rumor detection is essential to achieve this capability in the Digital Twin. Second, the component of data integration in the Digital Twin enables integrating the data from the first component to create knowledge graphs for a better understanding of disaster situations. However, the projection function in existing AI methods to reduce the dimensionality and redundancy of multimodal data is still in its infancy. Hence, developing effective projection functions to improve data integration and analytics is needed in future studies. The third component of the Digital Twin focuses on the coordination of actors for decision-making processes in disasters. Despite the advances of AI

techniques such as AlphaZero and Monte-Carlo tree search, modeling and analyzing the decision-making processes of relief actors remain challenging (Dwivedi et al., 2019). Future research can develop automated multi-agent models with deep learning capability to better learn and characterize actors' interactions in emergency responses. Finally, the Disaster City Digital Twin enables the examination of the dynamic networks of actors, tasks, information, resources, and infrastructure to improve the efficiency of disaster management and emergency response. Integrating AI methods such as Bayesian networks can lead to learning the influence of an entity on its connected entities and quantify their interactions to infer the synergetic operations. To improve this function of the Digital Twin, future studies can improve methods for prediction of dynamic network processes in disaster response and emergency response processes. Through addressing these challenges and opportunities, the adoption of the Disaster City Digital twin as the unifying paradigm can enable integrating various AI and ICT research streams related to disaster response and management.

7. Concluding remarks

This paper provides a vision for a Digital Twin paradigm to enable interdisciplinary convergence in the field of ICT and AI for disaster response and emergency management. The proposed paradigm is composed of four components: multi-data sensing for data collection, data integration and analytics, multi-actor game-theoretic decision making, and dynamic network analysis. The convergence of interdisciplinary research streams holds a strong promise for enhancing the performance of disaster management processes.

CRediT authorship contribution statement

Chao Fan: Conceptualization, Methodology, Visualization, Writing - original draft, Writing - review & editing. **Cheng Zhang:** Methodology, Writing - original draft. **Alex Yahja:** Methodology, Writing - original draft. **Ali Mostafavi:** Conceptualization, Supervision, Writing - original draft, Writing - review & editing.

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